بسم الله الرحمن الرحيم

نظري الماده م قبل الميدتيرم ودعواتكم بالتوفيق لينا جميعا

- **Signal Definition :**

It is a physical Quantity that has effect.

- **System Definition:**

It us a physical device that makes a few operations like filters.

- **Signal Processing:**

Any Operation that is processed on signal through the System.

- **Signal Processing Types & Classification:**

- Types:

1. Analog signal processing
2. Digital Signal processing

- Classifications:

1. Multi-channel – Multi Dimension.
2. Continuous / Discrete time.
3. Continuous / Discrete Value.
4 Deterministic / Random
5. Periodic / non-Periodic.

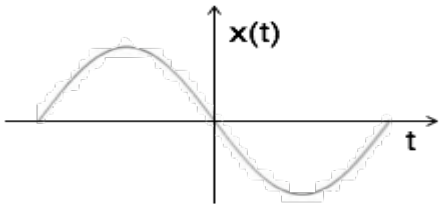
- **Advantages of Digital Signal Processing:**
- Flexibility.
- Enormous Storage Capacity.
- Controlled Accuracy

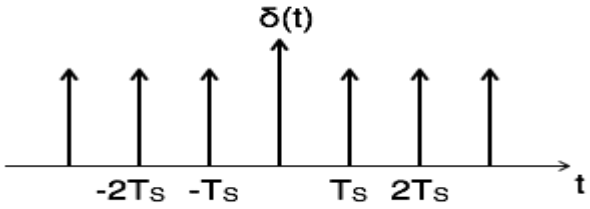
- **State and Explain sampling**

Sampling : Conversion of continuous Time signal to discrete time signal by taking samples .



Ts

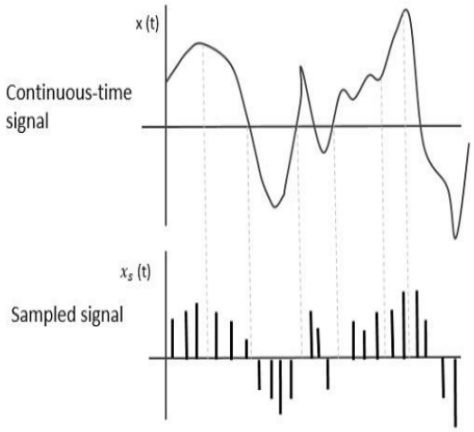




from nyquist Condition :

- Fs >=2Fmax
- Fs….Sampling Frequency
- Ts… Sampling Time
- Fs =1 / Ts (sample/sec)

- **Relation Between f , fs,F (Prove)**



Xa(t)=A cos (2πft+Θ)

t=nTs= N / Fs

Xa(t)=A cos(2πfnt+Θ)

= A cos(2πf N /Fs +Θ) …. (1)

Discrete signal: X(n)=A cos(2πFn+Φ) ...(2)

By Compare 1, 2: F= f / fs

Where: f ...frequency of cont. signal | F -> freq. of Disc. Signal

Fs ...Sampling Frequency.

- **proof Sampling condition (Nyquist).**

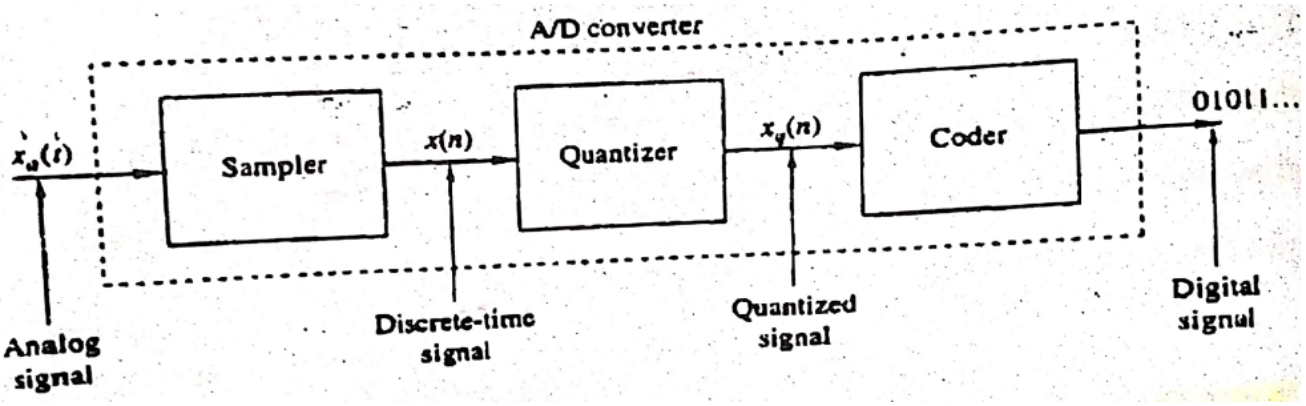
F≤ ½ cycle/sample.

F= f / fs

$f / fs \leq ½$

$f \leq fs\ 2 \rightarrow fs \geq 2f$ (Nyquist Condition).

- **Explain Basic 3 operation in A/D or Draw block diagram of Analog To digital Converter**



Sampling : Conversion of continuous Time signal to discrete time signal by taking samples .

Quantization : Done by Rounding or truncation of samples to discrete values/Levels ,causes error

Quantization Error : $-\Delta / 2 \leq eq \leq \Delta /2$

Coding : each quantized level represented in binary code

- **Continuous time VS discrete time**

Continuous time : is defined for every value of time and take values in continuous interval

discrete time : is defined only at a certain specific value of time

- **Continuous value VS discrete value**
- Continuous value : signal takes all possible values on a finite range
- discrete value: signal takes values from a finite set of possible values

- **Deterministic VS Random**

Deterministic : Signal can be represented by by a mathematically formula

Random : Signal can't be represented by by a mathematically formula

- **Multi-channel VS Multi-Dimensional**

Multi-channel : The signal generated by Multiple sources

Multi-Dimensional : signal Which function of independent variables

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- **proof x(k) is periodic**

$$x(k) = \sum_{n=0}^{N-1} x(n) e^{-jkn\frac{2\pi}{N}}$$

$$x(k+N) = \sum_{n=0}^{N-1} x(n) e^{-jn\frac{2\pi}{N}(k+N)}$$

$$x(k+N) = \sum_{n=0}^{N-1} x(n) e^{-jn\frac{2\pi}{N}k+(-j2\pi n)}$$

$$x(k+N) = \sum_{n=0}^{N-1} x(n) e^{-jkn\frac{2\pi}{N}} . e^{-j2\pi n}$$

$$e^{-j2\pi n} = 1$$

So x(k+N)=x(k) .. # periodic

- **Calculate Saving FFT - DFT OR (compute the computational saving of using FFT instead of DFT)**

	No. of Complex multiplication	N. of Complex addition
DFT	$N^2=16$	$N(N-1)=4*3=12$
FFT	$N/2 \log_2 N=4$	$N*\log_2 N=8$
Saving	16-4=12	12-8=4

#FFT Less operations than DFT – FFT faster than DFT

- **Interpolation Definition:**

Multiply discrete signal with sinc to return back to analog.

$$sinc\ x = \sin$$

متنساش تدعيلي وتدعي لزمايلك بالتوفيق ومتنساش الصلاه ع النبي عليه افضل الصلاه والسلام